DESIGN PROJECT

LLM Hallucination Detection using Multi-Layer EigenScore

PRESENTED BY

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1. Introduction

Large Language Models (LLMs) are increasingly used in education, search and decision support. However, they frequently **hallucinate**: they generate fluent, confident answers that are factually incorrect. Because a hallucinated answer is written in the same tone as a correct one, users cannot tell them apart by reading the output.

A reliable way to **estimate when a model is likely to be wrong** is therefore essential for deploying LLMs safely.

2. Problem Statement

> Design a system that, for any question posed to an LLM, produces a **hallucination score** indicating how likely the answer is to be incorrect, **without access to the correct answer** and **without external fact-checking**, while running on a **consumer GPU with 8 GB of memory**.

Existing approaches inspect only output word probabilities or the surface text of answers, and discard the rich semantic information that remains inside the model.

3. Objectives

- Detect hallucinations using the model's **internal hidden states**.
- Measure **semantic consistency** across multiple sampled answers.
- Reduce **over-confident** generations at inference time.
- Run a 7-billion-parameter model on an **8 GB GPU**.
- Validate the system quantitatively on a question-answering benchmark.

4. Key Idea

If a model knows the answer, repeated answers to the same question are **consistent in meaning**; if it is guessing, they **diverge**.

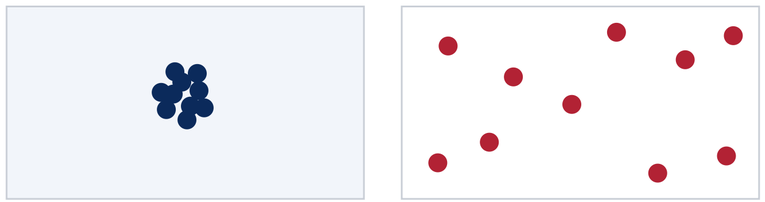


Fig. 1. Internal representations of 10 sampled answers (2-D projection, illustrative).

5. Methodology



Fig. 2. Processing pipeline of the proposed detector.

5.1 Semantic consistency score (EigenScore)

For a question, the LLM generates **K = 10** answers by sampling. For each answer we extract the hidden state of its last token, a 4096-dimensional vector z_i. The covariance matrix Σ of these vectors captures how the answers vary; its eigenvalues λ_i measure the spread along each direction.

$$E = (1/K) \cdot \log \det(\Sigma + \alpha I) = (1/K) \, \Sigma_i \log \lambda_i$$

$\alpha = 10^{-3}$ ensures numerical stability. A higher E indicates a likely hallucination.

5.2 Multi-layer aggregation

Rather than relying on a single, hand-picked layer, the score is computed over a **band of middle layers** (layers 8–24 of 32) and averaged, giving a more stable and model-independent signal:

$$E_{ML} = \text{mean}_{\ell \in [L/4,\, 3L/4]} \, E_\ell$$

5.3 Test-time feature clipping

A model that is confidently wrong repeats the same answer, so its answers appear consistent. To counter this, each neuron in the penultimate layer is **clipped to its 0.2–99.8 percentile range**, estimated from a memory bank of the last 3,000 tokens. This suppresses extreme activations that drive over-confident generation.

6. System Implementation

- **Model:** Llama-2-7B in **4-bit** precision (≈3.9 GB instead of 13.5 GB), PyTorch and HuggingFace Transformers.
- **Memory-efficient capture:** forward hooks store only the required layers instead of all hidden states.
- **Feature clipping module:** ring-buffer memory bank with per-neuron thresholds, applied during generation.
- **Verification:** 8 automated unit tests (all passing).

7. Experimental Setup

Parameter	Value
Dataset	CoQA (conversational QA), 250 questions
Samples per question	K = 10
Sampling	temperature 0.5, top-p 0.99
Correctness label	ROUGE-L > 0.5 vs. reference
Evaluation metric	AUROC
Hardware	NVIDIA RTX 4060 Laptop, 8 GB

8. Results

83.5% **AUROC for hallucination detection**

Probability that an incorrect answer receives a higher score than a correct one (50% = random).



Fig. 3. Detection performance on the 50–100% AUROC scale.

Table 1. Summary of results

Measure	Value
AUROC (EigenScore)	**83.5%**
AUROC (with feature clipping)	**82.5%**
Activations clipped	**0.29%**
Processing time per question	**≈ 6 s**
GPU memory for model weights	**≈ 3.9 GB**
Unit tests passed	**8 / 8**

9. Conclusion & Future Work

The system detects hallucinated answers from the model's internal states with an **AUROC of 83.5%**, needs no reference answers or external tools, and runs entirely on an 8 GB laptop GPU.

Ongoing: full evaluation of multi-layer aggregation, scale-normalised embeddings and a confidence-aware hybrid score on 1,000 questions each from CoQA, TriviaQA and Natural Questions.

10. References

[1] C. Chen et al., "INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection," ICLR, 2024.

[2] S. Reddy, D. Chen, C. D. Manning, "CoQA: A Conversational Question Answering Challenge," TACL, 2019.

[3] H. Touvron et al., "Llama 2: Open Foundation and Fine-Tuned Chat Models," arXiv:2307.09288, 2023.